Preprint

Measurement and sampling noise undermine inferences about awareness in location probability learning: A modeling approach

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Abstract

Occasionally, experimental psychologists enter into the realm of psychometrics without being fully aware of the risks involved in the study of individual differences. Here we re-assess the many studies on location probability learning that suggest that people can unconsciously learn to suppress salient but irrelevant distractors frequently presented in a certain location. In the additional singleton task, one of the arguments to support this claim is that suppression in search times does not significantly differ between "aware" and "unaware" participants. Although rarely acknowledged, this null interaction could also result if the data are contaminated by measurement and/or sampling noise. Unfortunately, the reliability of the awareness measure cannot be assessed with standard methods, since it is a single-trial test. In the present study we offer model-based estimations of measurement and sampling noise in empirical data. Our goal is to determine how often researchers would mistakenly conclude that learning is unconscious, given data from a model based on the opposite claim (i.e., that learning is conscious) but including noise in participants' search times and awareness responses. To do so, we fitted this noisy conscious model to a public dataset involving 159 participants who performed the additional singleton task. Estimated parameters from this model were used, first, to predict the observed pattern of results and, second, to simulate new responses and participants. Results suggest that, under reasonable measurement noise and sample sizes, simulated evidence from the model can paradoxically but falsely support arguments used to defend the unconscious learning hypothesis. This study serves as an illustration to experimental psychologists of the risks of neglecting basic psychometric requirements in individual differences research.

Key words: unconscious processes, awareness measures, location probability learning, measurement noise, sampling noise, cognitive modeling

Measurement and sampling noise can undermine inferences of awareness in location probability learning: A modeling approach

Psychometricians and experimental psychologists inhabit radically different worlds (Borsboom et al., 2009; Cronbach, 1957). The former explore domains where performance varies substantially across participants, looking for ever better ways to reliably detect those individual differences. The latter, in contrast, crave robust effects that, ideally, will be observed in every single person, often with little or no variation from one participant to the next. These two species rarely venture into each other's habitat and in the exceptional cases when this happens the outcome is not always a happy one.

Over the last decade, it has become increasingly common to use laboratory tasks developed by experimental psychologists to study differences across participants (Rouder & Haaf, 2019; Rouder et al., 2023). Unfortunately, although these tasks are known to yield large effects at the group level, they are typically insensitive to individual differences (Hedge et al., 2018). Consequently, attempts to use these tasks to study individual differences have sometimes resulted in contradictory findings and failed replications (e.g., Karr et al., 2018; Paap & Greenberg, 2013; Rey-Mermet et al., 2018; Ross et al., 2015; von Bastian et al., 2016). One potential reason for this is that, when experimenters enter into the realm of psychometrics, they commit to certain assumptions without evaluating whether their data validate them. A clear example consists in submitting variables to analyses that implicitly assume they have been measured with little or no measurement noise (Vadillo et al., 2020, 2022). On the occasions when this assumption has been evaluated, the psychometric properties of the measures provided by these tasks often leave much to be desired (e.g., Draheim et al., 2019; Enkavi et al., 2019; Garre-Frutos et al., 2024; Hedge et al., 2018; Paap & Sawi, 2016; Rothkirch et al., 2022; Vadillo et al., 2022; Yaron et al., 2024).

The study of unconscious learning and memory processes provides many examples of this practice. A common argument to support the claim that learning and memory can operate unconsciously is to report evidence that an improvement in task performance is uncorrelated with participants' awareness of the regularities driving performance (e.g., Colagiuri & Livesey, 2016; Jiang et al., 2018; Salvador et al., 2018; Soto et al., 2011). For instance, Soto et al. found a null correlation between performance in a working memory task and awareness of the cues' visibility, which served as an argument for the inference that working memory can operate with unconscious representations.

The intuition behind these analyses is that if participants' awareness and learning are uncorrelated, they must be based on different processes or representations. Learning must be based on something that leaves no trace on awareness measures. However, an alternative and far simpler explanation for the lack of correlation is that either or both measures are contaminated by substantial amounts of measurement noise. If this is indeed the case they will not correlate with each other, even if at the latent level they tap onto the same cognitive processes (Hunter & Schmidt, 2015; Spearman, 1904). One cannot discriminate between these two hypotheses (i.e., learning is unconscious vs. there is excessive measurement noise) without having some estimation of the internal consistency of the dependent variables involved in the analysis. Sadly, reliabilities are only seldom reported in these experiments (Vadillo et al., 2022) and, in the rare occasions when they are estimated, reliability often turns out to be sufficiently low to cast doubt on the appropriateness of the analysis (e.g., Anderson & Kim, 2019; Arnon, 2020; Bogaerts et al., 2018; Erickson et al., 2016; Kalra et al., 2019; Kaufman et al., 2010; Siegelman & Frost, 2015; Smyth & Shanks, 2008; Vadillo et al., 2020, 2022, 2023; West et al., 2018). To make things worse, several tasks used in the study of unconscious learning and memory apply this logic to single-item measures of awareness (such as probability cuing, e.g., Jiang et al., 2015; Vadillo et al., 2020; attentional capture, e.g., Adams & Gaspelin, 2020; and distractor suppression in the additional singleton task, Wang & Theeuwes, 2018a, 2018b), commonly taken at the end of the experiment. These designs are particularly challenging because the internal consistency of a single measurement is, by definition, undetermined, so reliability cannot be estimated with conventional methods (such as split-half reliability).

On another level, which is more extensively discussed in the literature, is the concept of sampling noise. When inferring a population effect through an empirical sample, the computed effect is only an estimation, which attempts to approximate the true population effect but is influenced by various uncertainties inherent in the sample. This is why participant sample size is one of the main considerations in designing experiments. Despite its importance, there is still no well-established routine to adequately justify sample sizes (Lakens, 2022), leading to samples that may not be sufficiently powered to detect the desired effects – in this case, an interaction between awareness and learning. A common argument to justify testing a specific number of participants is that the sample size is similar to previous ones in the same literature, but this does not guarantee that those sample sizes were sufficiently informative. For instance, in the domain of location probability learning, which is the main focus of the present study, median sample

sizes are typically between 16 and 24[1]. With these samples, the minimum detectable correlation with 90% power is .705 for contextual and probabilistic cuing tasks and .601 for the additional singleton task[2]. Given that some amount of measurement noise is a very reasonable assumption in the measures taken in these tasks, there are good reasons to expect that the empirical correlations involving them will often be substantially smaller than these values.

In the present study we illustrate the scope of these problems – measurement and sampling noise – in the domain of location probability learning by adopting a modeling-based approach. This approach will allow us to achieve two goals: First, to provide a model-based estimate of how much measurement and sampling noise there is in empirical data from these tasks; second, to find out how often researchers would mistakenly conclude that learning is unconscious, given a model based on the alternative assumption (learning is conscious) but varying the amounts of measurement and sampling noise in participants' responses. In sum, this study serves as a timely illustration for experimental psychologists on the risks of neglecting basic psychometric requirements when assessing individual differences.

Location probability learning in the additional singleton task

Numerous studies have argued that our ability to guide attention is strongly influenced by unconscious learning processes that enable us to detect and exploit statistical regularities in our environment (Chun & Jiang, 1998; Chun & Turk-Browne, 2008; Ferrante et al., 2018; Geng & Berhmann, 2005; Jiang, 2018; Gaspelin & Luck, 2018; Krishnan et al., 2022; Stilwell et al., 2019; Turk-Browne et al., 2005). While much of this research is concerned with how we learn to attend to locations where a target is likely to appear, many studies have made the complementary claim that with sufficient practice, we can unconsciously learn to ignore locations in a scene where salient but irrelevant distractors typically appear. This phenomenon is commonly studied using the additional singleton task (e.g., Di Caro et al., 2019; Gao & Theeuwes, 2020, 2022; Lin et al., 2021; van Moorselaar & Theeuwes, 2021; Wang & Theeuwes, 2018a, 2018b). This increasingly popular experimental task serves as a perfect example for the present purposes.

In a typical setting, participants are instructed to find a target with a particular shape among a series of shapes (e.g., a diamond among circles) and report the orientation of the line inside the

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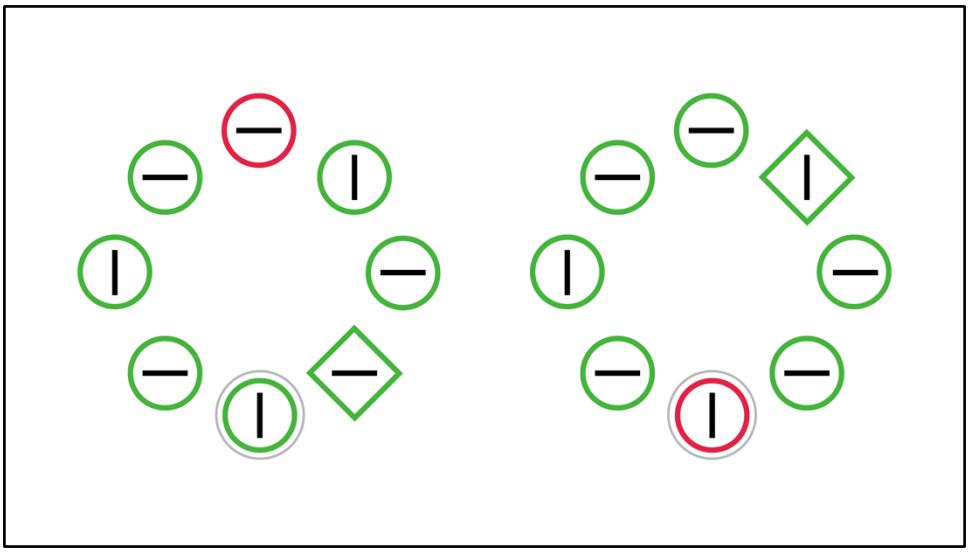
[1] Median $N = 16$ for contextual cueing and probability cueing and median $N = 24$ for the additional singleton task (Vadillo et al., 2016, 2020; Vicente-Conesa et al., 2022).

[2] These power calculations were obtained with the function `pwr.cor.test()` from the R package `{pwr}` (Champely, 2020).

target (see Figure 1). Participants are warned that in some trials the display includes a salient distractor presented in a different color (e.g., a red circle among green figures). Response times (RTs) reveal that visual search is slower when this singleton is present (e.g., Theeuwes, 1992). Presumably this happens because participants engage a strategy to detect any stimulus with a unique feature, even if this leads them to pay attention to irrelevant stimuli as well (i.e., singleton detection mode; Adams & Gaspelin, 2020; Bacon & Egeth, 1994). Crucially, if the singleton distractor appears frequently at a particular location throughout the experiment, participants are able to detect this regularity and suppress attention to that particular location (e.g., Vicente-Conesa et al., 2022; Wang & Theeuwes, 2018a, 2018b). This suppression effect is shown by the fact that, as learning progresses, search times for the target are faster when the salient distractor appears at the high-probability (HP) location compared to when it appears at any other low-probability (LP) location. In essence, participants learn not to be distracted by a salient stimulus if it appears frequently at a given location. Figure 1 illustrates two possible displays of the additional singleton task for one participant.

Figure 1

Illustration of two trials of the additional singleton task



Note. The target is a distinct shape (diamond) and the singleton distractor is a salient color (red). The high-probability (HP) location is indicated with a grey circle (not shown on the actual experimental trial). In the left display, the distractor appears at a low-probability (LP) location while in the right display it appears at the HP location.

At the end of these experiments, participants are informed that the distractor appeared more frequently at a certain location and then they are asked to identify and report this location (i.e.,

the awareness test). Although the suppression effect is very robust in RTs, many participants seem to be unable to correctly select the HP location (e.g., van Moorselaar & Theeuwes, 2021). When this happens, authors infer that the location probability learning was implicit. In other studies, researchers remove those participants who correctly selected the HP location – the *aware* ones[3] – from analyses and observe that the suppression effect is still significant in the remainder (e.g., Di Caro et al., 2019). Again, this is taken as evidence that learning was unconscious, in the sense that awareness is not necessary for the effect to occur.

A third strategy consists of dividing the sample between those participants who correctly selected the HP location and those who did not – the *aware* and the *unaware* ones – and comparing their suppression effects. This is usually performed through a two-way mixed ANOVA for RTs with one within-subjects factor that assesses suppression, the distractor location (HP or LP), and one between-subjects factor, accuracy in the awareness test (aware or unaware participants)[4]. Obtaining a non-significant interaction between these factors has been typically interpreted as evidence that location probability learning is unconscious (e.g., Failing et al., 2019; Gao & Theeuwes, 2022; Lin et al., 2021). In other words, if the distractor suppression effect, as measured in RTs, is not related to correctly responding to the awareness test, this is taken as evidence that the suppression effect and awareness must be generated by unrelated (or weakly related) latent processes (as Rouder & Haaf, 2019, suggest for inhibition). At first glance, the logic of these analyses seems completely sensible. However, somehow implicitly, we have moved from experimental logic and have entered the realm of individual differences. It is here where psychometric properties reign and, therefore, we need to address them to guarantee valid inferences.

Risks of blindly following this logic

An astute reader may have noticed that claiming support for a theory based on a non-significant result is scientifically unsound: As Carl Sagan (1977) said, “absence of evidence is not evidence of absence” (p. 7). Yes, a non-significant interaction can be interpreted as evidence for the claim that location probability learning is independent from awareness but, have we

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[3] Prior to this measurement a confidence rating is often taken on the participant’s explicit awareness of the manipulation – whether they were aware that the singleton distractor appeared more probably at a certain location. Some authors classify participants as unaware not only on the basis of being unable to identify the HP location but also based on reporting no conscious awareness in their rating.

[4] Note that this interaction is equivalent to observing a simple difference in the magnitude of the suppression effect between aware and unaware participants.

discounted alternative plausible scenarios in which a non-significant interaction was expected to occur? In other words, let's put ourselves in the opposite scenario and imagine that RTs and awareness are driven by a common underlying process. Would we have any reason to expect a non-significant interaction in that case too?

Imagine we are omniscient and know that there is a true interaction effect, that is, the true suppression effect in aware participants is indeed greater than in unaware ones. From our privileged point of view, the correct claim would be that "location probability learning is conscious" or, more precisely, that there is a true relationship between distractor suppression and awareness. Now imagine two research teams that want to check this interaction with empirical data. The first team has collected data and is preparing to analyze them. Unfortunately, these researchers encountered difficulty in recruiting participants and, although they completed a large amount of trials, the study ends up with a small sample size. As a consequence of their lack of statistical power to detect the effect, they obtain an empirically non-significant interaction. The second research team, with much better access to participants, has obtained their measures with a sufficient sample size. However, these participants had less time available and so only completed a small number of trials, increasing the measurement noise of their dependent variables. As has been well known for decades now, measurement noise tends to attenuate true effects (Hunter & Schmidt, 2014; Lord & Novick, 1968). Unfortunately, these researchers' measurement noise was sufficient to attenuate the actual interaction effect to the point that they also obtain an empirically non-significant interaction.

What claim might both teams risk drawing from those non-significant interactions? That the suppression effect is unrelated to awareness responses and, thus, that there is evidence supporting the claim that location probability learning is unconscious? From our omniscient perspective, we know they will be making an erroneous claim and can alert them about the need to first eliminate both sources of attenuation affecting the design (insufficient sample size: sampling noise) and measures (insufficient reliability: measurement noise).

Goals of the present study

Unfortunately we are not omniscient, but we can adopt the privileged position of data simulation where the ground truth is known. To simulate data, we first need a model of how our dependent variables are distributed as a function of the independent variables manipulated in the experiment. In this article, we present a statistical model to describe both RTs and awareness

responses in each participant in the additional singleton task. Given the handicaps of estimating the reliability in experimental data (impossible for the single-trial awareness measure), the model allows us to simulate data from a single-process theory while varying sampling and measurement noise.

This modeling exercise will serve as an informative illustration of how neglecting basic psychometric requirements, such as sufficient sample size and minimal measurement noise, can undermine the statistical power to detect effects of interest. Note that our goal is not to find the most accurate underlying model for real cognitive processes involved in the additional singleton task nor in location probability learning. Instead, what we will try to show is how an empirical result often found in the literature, and interpreted as evidence for a certain theory, can be generated from a contrasting theory plus some statistical artifacts. Therefore, our aim is to encourage future researchers, working on this and related tasks and research questions, to meticulously assess whether their data fulfill the crucial requirements for drawing valid conclusions about individual differences in the experimental context.

Modeling visual search and awareness in the additional singleton task

Before entering into the details of the model, there is a general consideration we need to point out. We are interested in modeling how the experimental manipulation applied in the additional singleton task (i.e., that the distractor appears more frequently at a specific location) affects both RTs and awareness responses. In the previous literature, this influence is captured (1) as the suppression effect in RTs and (2) as accuracy in the awareness test. For the former, this influence is commonly calculated as the RT difference between LP and HP distractor conditions. Note that this specific score treats trials where the distractor appeared *close* to the HP location as equivalent to those where the distractor appears *far* from it. Similarly, for the awareness measure, authors typically divide the sample into two groups based on their awareness response: participants who correctly selected the HP location and those who failed to do so (e.g., Failing et al., 2019; Gao & Theeuwes, 2022; Lin et al., 2021). Again, note that doing this treats participants who selected a location *close* to the HP location as equivalent to those who selected a location *far* from it. In summary, both variables are dichotomized before analyzing the interaction between them.

This double dichotomization would be unproblematic if there were no evidence of a spatial gradient effect. However, many studies have found evidence for such a spatial gradient, both for

RTs (Gao & Theeuwes, 2020; Failing et al., 2019; Vicente-Conesa et al., 2022; Wang et al., 2019; Wang & Theeuwes, 2018a, 2018b) and awareness responses (Vicente-Conesa et al., 2022). For instance, a typical result is that the effect of including a singleton in the search display is progressively reduced the closer it is to the HP location. Similarly, RTs to find a target progressively increase the closer it is to the HP location. Finally, in awareness tests, even when participants make a mistake, they most commonly choose locations close to the HP location. Consequently, any model for data from this task should ideally incorporate this spatial aspect, accounting for differences between distances to the HP location as a ordinal variable instead of a binary one. Below the reader will see that predictors in our model are chosen because they represent these distances and can account for the spatial gradient.

Modeling visual search

We modeled RTs in the visual search task with an ex-Gaussian distribution, which results from convolving a normal and an exponential distribution. Thus, the model is $RT \sim$ ex-Gaussian $(\mu_{RT}, \sigma_{RT}, \tau_{RT})$, with μ_{RT} and σ_{RT} being the mean and standard deviation of the normal component of the distribution, respectively, and τ_{RT} the rate parameter of the exponential component (Luce, 1986). Previous research indicates that the ex-Gaussian distribution provides an excellent fit to RTs in a wide variety of visual search tasks (Palmer et al., 2011).

Our model assumes that, for each participant, three specific parameters, μ_{RT}, σ_{RT}, and τ_{RT}, describe their RTs on each trial. However, not every display condition will evoke the same RT distribution. For the sake of simplicity, our model predicts that changes in the properties of each display will affect the parameter μ_{RT}, but not σ_{RT} and τ_{RT}, which will remain constant for each participant[5] (for a similar approach, see Rouder & Haaf, 2019). So, what properties, manipulated in each display of the additional singleton task, can cause variations in the μ_{RT} parameter?

After evaluating alternative models with different variables and parameters (see the Supplementary Material), we finally selected a model which includes two predictors based on the

[5] We tested if this assumption was reasonable estimating the three parameters in both HP and LP distractor conditions separately and comparing each pair of parameters with a repeated-measures *t*-test ($\alpha = .05$). This analysis indicates that μ_{RT} is the parameter with the largest difference across conditions ($t_{(158)} = 22.955$, $p < .0001$, $d = 1.82$ with 95% CI [1.57, 2.07]), much larger than σ_{RT} ($t_{(158)} = 6.592$, $p < .0001$, $d = 0.52$ with 95% CI [0.36, 0.69]) and τ_{RT} ($t_{(158)} = -10.922$, $p < .0001$, $d = -0.87$ with 95% CI [-1.05, -0.68]), although these two also reached significance. The R script for these analyses is available at https://osf.io/baxf8/.

two most influential features of the search display: the distance from the target to the HP location (*dT*) and the distance from the distractor to the HP location (*dD*), in each trial. In Figure 1, two displays with different *dT* and *dD* are shown to illustrate the meaning of these parameters. In the left display, *dT* is 1 and *dD* is 4 while in the right display, *dT* is 3 and *dD* is 0. Note that, if the distractor has a distance of zero to the HP location (grey circle, *dD* = 0), this designates a HP trial (e.g., the right display).

Previous studies show how these distances affect the μ_{RT} of participants that, after some trials, have learnt (consciously or unconsciously) which is the HP location (e.g., Wang & Theeuwes, 2018a, 2018b). On trials where the target appears further from the HP location, they are expected to take less time to find the target, consistent with the idea of a gradient of suppression around the HP location. Note that the model suppresses unwanted interference from a distractor and inhibits wanted processing of the target via the same process. On trials where the distractor appears further from the HP location, participants are expected to take more time to find the target, because the distractor is not being suppressed. Separately, these influences can be mathematically expressed as the following simple linear regressions:

$$\mu_{RT} = \beta_0 - \beta_1 \cdot dT \quad [1]$$

$$\mu_{RT} = \beta_2 + \beta_1 \cdot dD \quad [2]$$

In these equations, β_0 and β_2 are the respective intercepts of each regression, the parameters for the visual search fixed cost time when *dT* or *dD* equal 0. β_0 can be conceptualized as a general visual search cost time, while β_2 is a singleton-presence cost time. The slope β_1 is the most relevant parameter in both equations. It represents the weight of *dT* and *dD*; that is, the change in μ_{RT} as the distance from the target or distractor to the HP location increases. Imagine a participant who has failed to learn that the distractor appears more frequently at a certain location. In our model, this would translate to a β_1 close to zero. Consequently, distances from target or distractor to the HP location would not affect μ_{RT}. The theoretical implication of β_1 is important in the study of location probability learning because it indicates the degree to which participants *learn* where the HP location is and create an attentional strategy to suppress it. This leads us to assign a negative sign for *dT* and a positive one for *dD*: A shorter target distance and a larger distractor distance result in a slower target detection by the participant; thus, a higher μ_{RT} is expected in those trials.

In the Supplementary Material, readers will find technical details explaining why fixing the slope parameter, β_1, to the same value in both regressions does not result in a significant worsening in model fit. The same reasoning motivated us to fix to zero the slope of a third possible but discarded predictor, the target–distractor distance (*dTD*; van Moorselaar & Theeuwes, 2021), which was found to have little influence on μ_{RT} (also presented in the Supplementary Material). In summary, the complete model for RTs can be expressed as

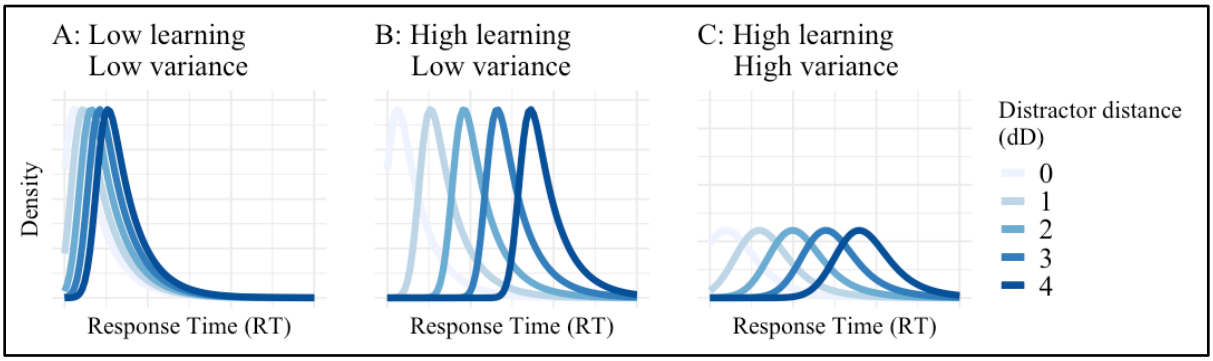
$$RT \sim \text{ex-Gaussian}\left(\mu_{RT} = [\beta_0 - \beta_1 \cdot dT + x\left(\beta_2 + \beta_1 \cdot dD\right)], \sigma_{RT}, \tau_{RT}\right), \qquad [3]$$

where x is a switch variable which takes the value 1 in trials with a distractor and the value 0 in trials without a distractor. This way, we can control when activating the part of the formula that involves *dD*. The model assumes that the distance from the target to the HP location, *dT*, is important on all trials, even when no distractor is actually presented.

How measurement noise in RTs is modeled. According to the model, the process we want to measure (i.e., location probability learning, represented by β_1) is included only in μ_{RT}. Every other influence affecting the distribution can be considered a source of uncertainty, which is here represented by the ex-Gaussian variability parameters, σ_{RT} and τ_{RT}, or more concisely by its variance (calculated as $\sigma_{RT}^2 + \tau_{RT}^2$). The larger σ_{RT} and τ_{RT} are, the more the RTs will tend to deviate from the predicted μ_{RT} for the corresponding experimental condition. In this sense, these parameters can be understood as a source of noise that on average distances observations from the values that would be expected based on the latent process responsible for the experimental effects. The most significant effect of this measurement noise is to attenuate the size of the manipulation effect. Figure 2 illustrates this attenuation effect. A participant with a low learning parameter (Figure 2A) is expected to show small differences as *dD* varies. As this parameter increases (Figure 2B), ex-Gaussian curves overlap less. However, if these RTs are measured with excessive measurement noise (i.e., increasingly larger values of σ_{RT} and τ_{RT}), the distributions will be flattened (Figure 2C).

Figure 2

Illustration of modeled distribution for response times (RTs) in five different trial conditions each with a specific distance from distractor to high-probability location (dD), as a function of variance in RTs



Note. These ex-Gaussian curves do not follow the complete model presented in this article, but a reduced example using only one predictor: the distance from distractor to high-probability location (*dD*).

Modeling awareness

Our model assumes a single learning process, that is, the same underlying mechanism that determines RTs as a function of target and distractor distances to the HP location also determines the response in the awareness test. Specifically, the model states that when participants are explicitly asked to identify the HP location, their awareness response, AW, is extracted from a multinomial distribution with the eight locations as categories:

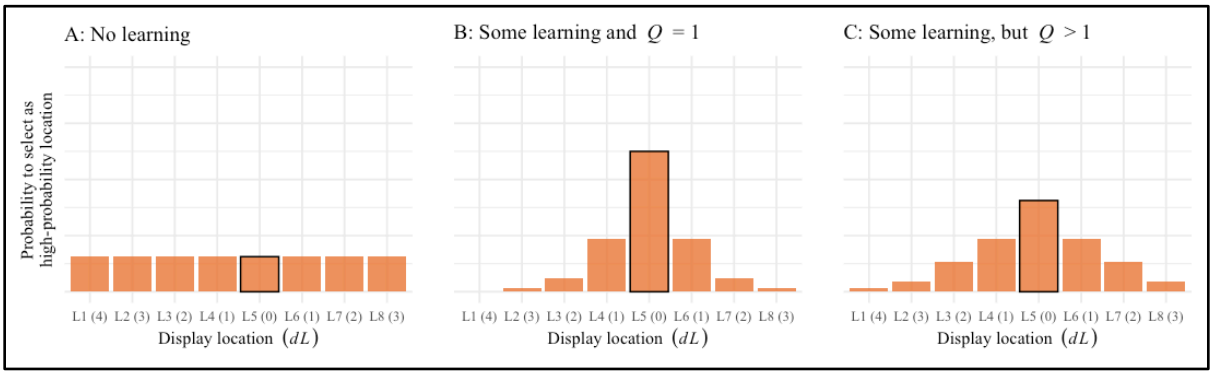
$$AW \sim \text{Multinomial}\left(1, p_l = \frac{e^{w_l/Q}}{\sum_{i=1}^{L} e^{w_i/Q}}\right), \text{where } w_l = -\beta_1 \cdot dL_l, \text{ for each } l \in \{1,...,8\}, \quad [4]$$

where p_l is the probability of selecting location l as the HP location in the awareness test at the end of the experiment. This probability is calculated with the SoftMax rule, $\frac{e^{w_l/Q}}{\sum_{i=1}^{L} e^{w_i/Q}}$, a function that transforms a vector of numbers (the eight locations of the display) into probabilities ranging from 0 to 1, whose sum equals 1. These numbers are transformed into probabilities, w_l, to reflect the influence of the participant's learning, β_1, at the time of the awareness test, as a function of the distance, dL_l, of a location l to the HP location. Consequently, w_l is the result of another simple linear regression: $-\beta_1 \cdot dL_l$. Indeed, this β_1 is the same parameter from the RT equation [3]. The parameter Q will be explained later.

Imagine a participant who has not learnt anything about where the HP location was across trials. As explained above, their parameter β_1 would be close to zero. In that case, at the time of responding to the awareness test, the distance from each location to the HP location, dL_l, would not affect their probability of selecting any location in particular and every w_l would be close to 0. In other words, they would have equal probability of selecting each location as the HP one (Figure 3A). Conversely, for a participant demonstrating greater learning regarding the HP location, the parameter β_1 increases, and the influence of dL_l becomes stronger. Specifically, locations with shorter dL_l should have greater probabilities of being selected in the awareness test, thus the negative sign in the equation for w_l (Figure 3B).

Figure 3

Illustration of modeled location selection distributions in the awareness test



Note. Panel A represents no learning about the high-probability location (L5, with black border). Panels B and C represent successful location learning, in B with a temperature parameter (Q) of 1, and in C with a Q higher than 1.

How measurement noise in AW is modeled. In Figure 3B, the influence of learning (β_1) on the multinomial distribution from which the awareness response is extracted will be most pronounced when the temperature parameter, Q, equals 1. Figure 3C represents the same learning as in Figure 3B, but with a higher value of Q. Note that this temperature parameter attenuates the influence of learning on the awareness response, increasing the probability of choosing any other (incorrect) location instead. Similar to RTs, the temperature parameter allows us to model the amount of measurement noise in the awareness response variable.

In the following section, we present two modeling studies. First, we fit the visual search and awareness model to real data from the additional singleton task (Vicente-Conesa et al., 2022). We show that the model predicts most of the trends observed in the data, including a very small interaction between learning and awareness that often will fail to reach statistical significance in

empirical studies with small and unreliable samples. Next, we report the results of a simulation assessing how statistical power to detect this interaction between learning and awareness can be expected to change as a function of sample size and measurement noise in both dependent variables (RTs and AW).

Fitting the model to observed data

Description of the dataset

The data used to fit the model were collected by Vicente-Conesa et al. (2022). The participants were recruited in three separate experiments of 80 each, but as explained below we only modeled data from Experiments 1 and 2. It is worth noting that these sample sizes are much larger than the typical ones found in this literature (a median of 24 participants from all studies reviewed in the Supplemental Material of Vicente-Conesa et al., 2022). Furthermore, data were collected and analyzed following a preregistered protocol. The visual search task was closely based on previous studies (Wang & Theeuwes, 2018a, 2018b) and the resulting suppression effect was very robust. Although each participant completed 720 visual search trials, the number of valid trials for each participant – after applying the same exclusion criteria as in the original study – ranged from 585 to 685 with a mean of 661 trials. The visual search and awareness model presented in the introduction was implemented in the R language (Version 4.3.1, R Core Team, 2023) and all scripts to reproduce our results are available at https://osf.io/y8v74/.

Each experiment varied only in the measurement method for the awareness test at the end of the task. In Experiment 1 participants were asked to identify the HP location while in Experiment 2 they ranked the three locations that in their opinion had contained the singleton distractor most frequently. It is relatively easy to fit our model to these two dependent measures (ignoring locations ranked second and third in Experiment 2) and, accordingly, we used data from both of them. In Experiment 3, participants estimated how many times the distractor appeared at each location, thus, were not asked directly to select the HP location. Therefore, we excluded this experiment and our dataset contains RTs and AW from 159 participants (data from one participant were excluded due to a technical problem).

Parameter estimation

We estimated the five parameters (β_0, β_1, β_2, σ_{RT}, and τ_{RT}) of the model separately for each of the 159 participants. We searched for parameters that maximize the log-likelihood of the

model using the `optim` function from the `{stats}` package in R (R Core Team, 2023) and the Nelder and Mead (1965) algorithm, as recommended by Balota et al. (2008). We fitted the basic ex-Gaussian distribution to all trials in each participant (without any predictors) and selected the resulting parameters μ , σ, and τ as starting values to estimate parameters β_0, σ_{RT}, and τ_{RT} in the model, respectively. The starting values for the remaining parameters, β_1 and β_2, were fixed to zero. R scripts and data for these estimations are available at https://osf.io/y85w2/.

Note that the temperature parameter, Q, cannot be estimated at the participant level because there is only one observation (the single-item awareness measure) per participant. To obtain a plausible estimate of Q, we first estimated the rest of the parameters with Q fixed to 1. Then, for each participant, we analytically calculated the probabilities of selecting each location of the display in the awareness test, using the equation [4] of the model. This procedure was repeated by increasing the Q parameter (common to all participants) until we found the value of Q for which the multinomial distribution of probabilities across participants best reproduced the observed distribution of awareness responses in Vicente-Conesa et al. (2022).

We analytically computed the predicted RTs for each trial as the mean $\mu_{RT} + \tau_{RT}$ resulting from applying *dT* and *dD* of the corresponding trial display and the estimated parameters of the corresponding participant. For the awareness response in each participant, we obtained the probability of each dL_l resulting from using the SoftMax rule in the model's equation [4].

Assessing model fit

Table 1 presents the descriptive statistics for each parameter in Vicente-Conesa et al.'s (2022) sample of 159 participants. As can be seen in Figure 4, at the group level, the model provides an adequate fit to the mean RTs across distractor conditions, no-distractor (ND), HP, and LP (Figure 4A), distractor distances (*dD* = [0...4]) in distractor-present trials (Figure 4C), and target distances (*dT* = [0...4]) in distractor-absent trials (Figure 4E). Note that, for all conditions, the 95% CI of the predicted mean (dots) overlaps with the 95% CI of the observed mean (bars), suggesting that the means can be considered equal (with α = .05). It is possible that the greatest discrepancies between predicted and observed mean RTs (i.e., conditions *dD* = 2 and 3 in Figure 4C, and *dT* = 1 and 2 in Figure 4E) are due to a non-linear tendency underlying the distances. For the sake of simplicity, we used a linear function for both predictors (*dD* and *dT*), but other alternatives, such as an exponential function, could be worth exploring in future models. At the participant level, the model also performs adequately as illustrated in the right figures (4B, 4D,

and 4F), since the dots are systematically distributed around the diagonals of the observed-predicted scatterplots.

Table 1

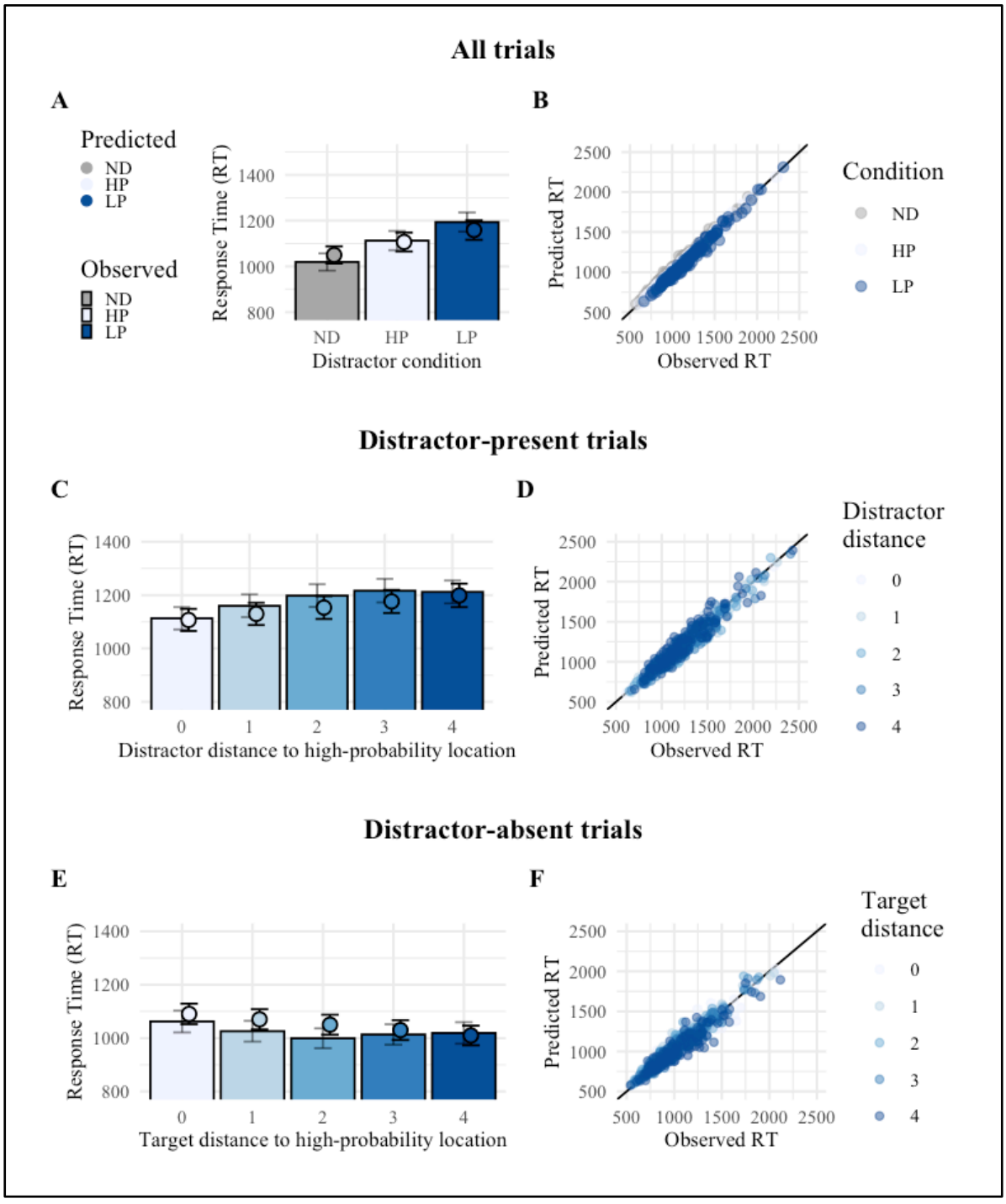
Descriptive statistics for each parameter for the data by Vicente-Conesa et al. (2022)

Parameter	Mean	Standard deviation	Min	Max
β_0	701.79	140.95	444.62	1337.61
β_1	20.14	11.58	-36.54	56.40
β_2	62.63	43.81	-22.02	265.00
σ_{RT}	102.40	50.11	31.26	306.86
τ_{RT}	388.79	141.85	159.94	1032.12

Although Figure 4 represents the trends in the data typically analyzed in this literature (e.g., corresponding to Vicente-Conesa et al., 2022, Figures 2 and 3) and our model reproduces the observed results at that level of analysis, the fit of the model can also be assessed at a finer level of granularity. Figure 5 illustrates the model fit of mean RTs in each of the 28 possible display conditions (i.e., 28 combinations between possible values of *dT* and *dD*). Here we can appreciate the caveats of the model predictions, especially when *dT* equals 0, that is, when the target appears at the HP location. Although the model predicts a pattern of increasing RTs as *dD* increases (common to every *dT*), the observed data show the opposite pattern: participants find the target *faster* as the distractor is located further from the HP location, where the target is located. The model would predict this effect if we had a term for the distance between target and distractor (*dTD*). As noted above, we excluded this third predictor on the basis that adding it did not significantly improve the model fit (see the Supplementary Material). Future models focusing on precise qualitative aspects of performance (van Moorselaar & Theeuwes, 2021) could include this predictor.

Figure 4

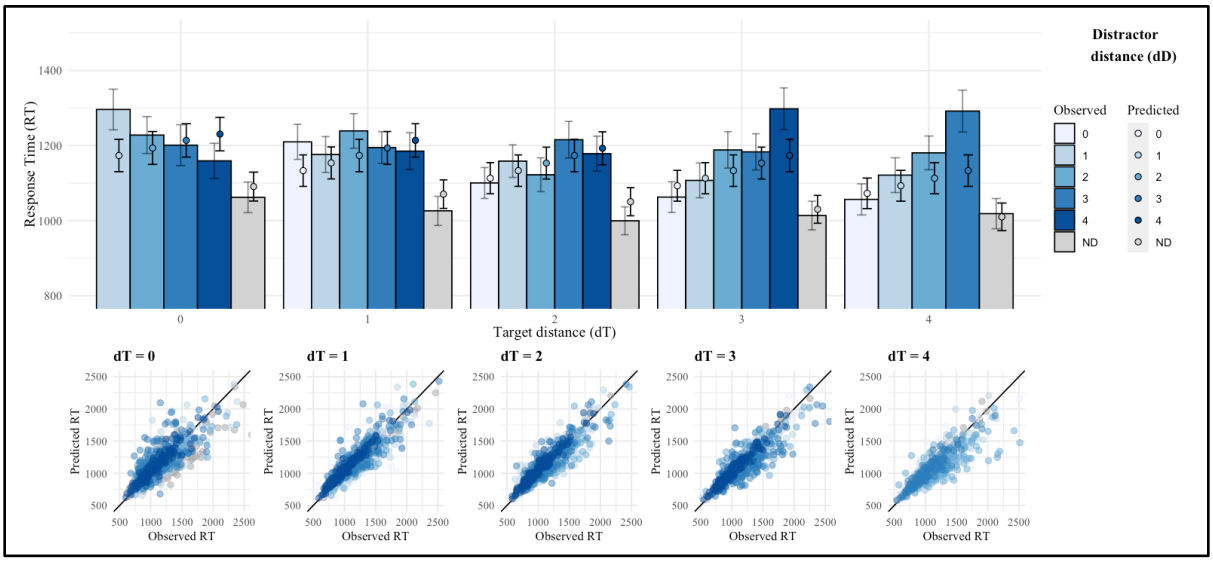
Response times (RTs) compared between observed (bars) and predicted data (dots), across distractor conditions (first row), distractor distances (second row), and target distances (third row). Figures on the right represent the fit between observed and predicted mean RTs for each participant in each condition



Note. ND: No-distractor trial; HP: High-probability trial; LP: Low-probability trial. Panel A represents the same data as in the left panel of Figure 2 in Vicente-Conesa et al. (2022, p. 801). Panels C and E represent the same data as in panels A and C, respectively, of Figure 3 in Vicente-Conesa et al. (p. 802).

Figure 5

Top panel: Comparison between observed (bars) and predicted (dots) response times (RTs), across dT (distance from target to high-probability location) and dD (distance from distractor to high-probability location) conditions. Bottom panel: Fit between observed and predicted mean RTs for each participant in each dT and dD condition

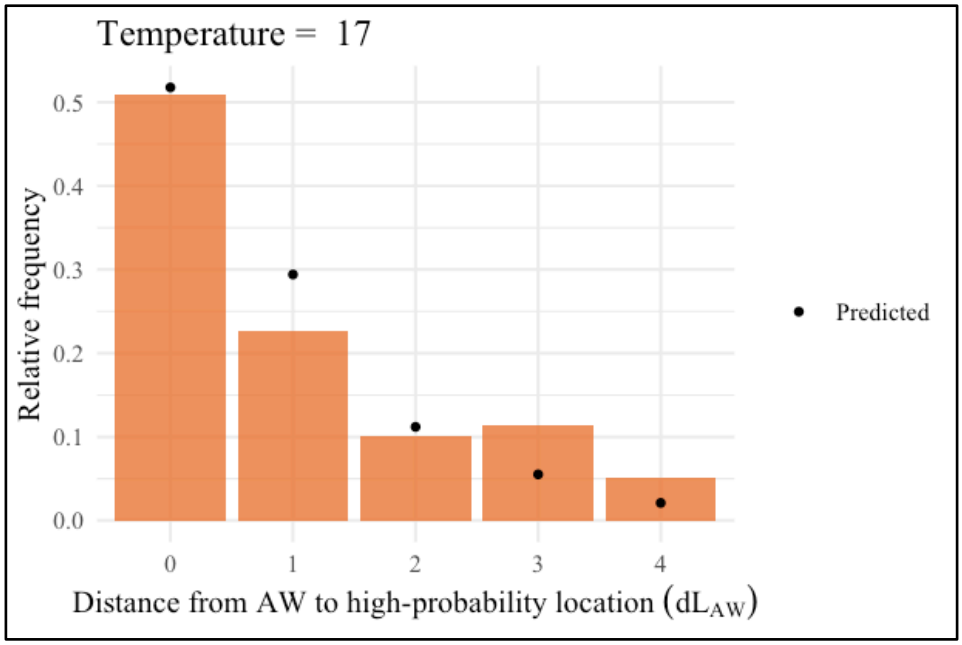


Note. ND: no-distractor trial. Due to differences in trial numbers in each condition and participant, reported observed RTs are the fixed intercepts from an intercepts-only mixed model for each condition (Level 1) and participant (Level 2), which isolates the within-participants variability. The model predicts the same RT for every trial from the same condition and participant, so the dots are the mean across participant predictions in each condition (one per participant).

Finally, Figure 6 illustrates the observed and predicted performance in the awareness test, specifically, how frequently participants selected locations at each distance from the HP location (dL_{AW}) in the observed data (bars) and the average probability of each response according to the model (dots). A temperature of 17 was the value which generates the predicted dL_{AW} probabilities most similar to the observed relative frequencies, reducing average error between them to 0.175 in proportion units.

Figure 6

Comparison of relative frequencies of selected awareness response (AW) in observed (bars) and predicted (dots) data, as a function of their distances from this response to the high-probability location (dL_{AW})



Detecting the interaction between suppression and awareness

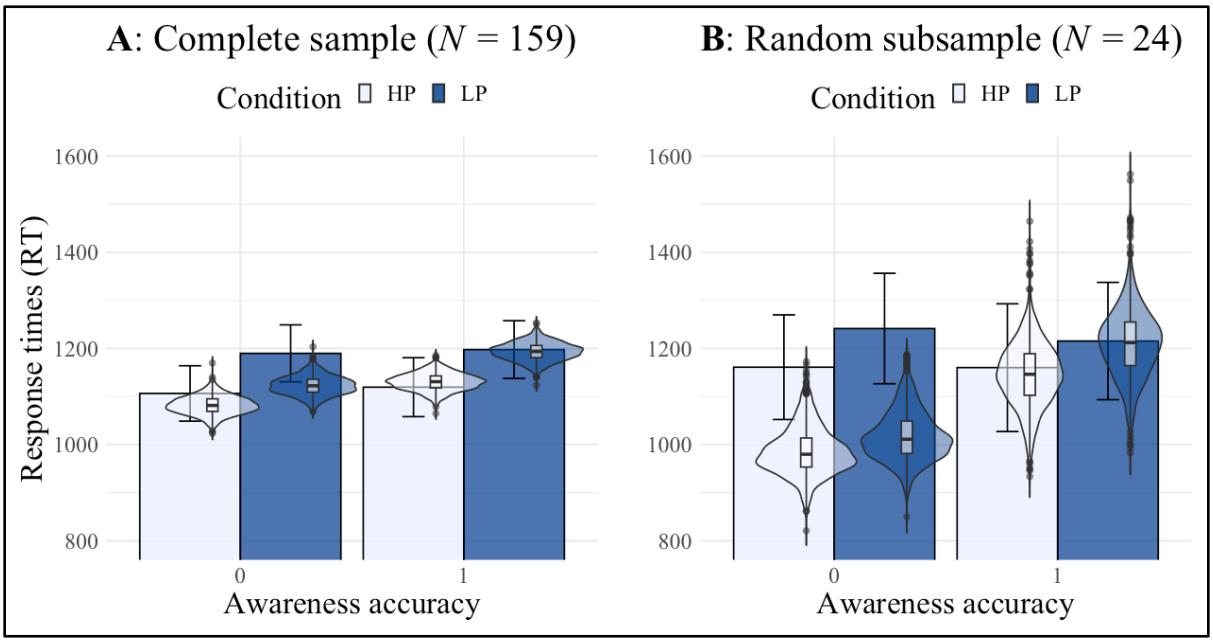
Researchers using the additional singleton task or similar paradigms tend to obtain a non-significant interaction between suppression (measured by the difference in RTs between HP and LP conditions) and awareness (whether the participant correctly selects the HP location, *aware*, or not, *unaware*). When analyzing Vicente-Conesa et al.'s (2022) dataset, we obtained the same non-significant interaction, $F_{(1, 157)} = 0.45$, $p = .501$, $\eta_p^2 < .001$. This result has typically led researchers to infer that learning is unconscious, neglecting to test whether the non-significance is compatible with a true latent interaction that has been attenuated by an excessive sampling and/or measurement noise. In the following section we simulate data according to the model described above. The model assumes a true interaction because learning is characterized by a single latent process, represented by the parameter β_1, which affects both RTs and awareness. Hence this simulation will illustrate the statistical power to detect this interaction under reasonable sampling and measurement noise.

In the previous section, we described the model's ability to predict search times and awareness by analytically calculating the mean RTs and probabilities of choosing each location as AW. These predictions are noiseless since there is a unique value for each condition and

participant. For the following illustration, we more appropriately simulate data with trial noise, randomly sampling each RT or AW from its corresponding distribution (the ex-Gaussian for RTs and the multinomial for AW). Figure 7 shows the mean predicted effects across 1,000 simulated samples (A) with the same design properties (trial conditions, and numbers of trials and participants) as in Vicente-Conesa et al.'s (2022) dataset, and (B) with a random subsample of 24 participants.

Figure 7

Interaction effect in response times (RTs) between trial condition (high-probability, HP, and low-probability, LP) and awareness accuracy (0, wrongly selecting the HP location, and 1, correctly selecting the HP location). Bars represent the observed data by Vicente-Conesa et al. (2022) and the violins the distribution of simulated results



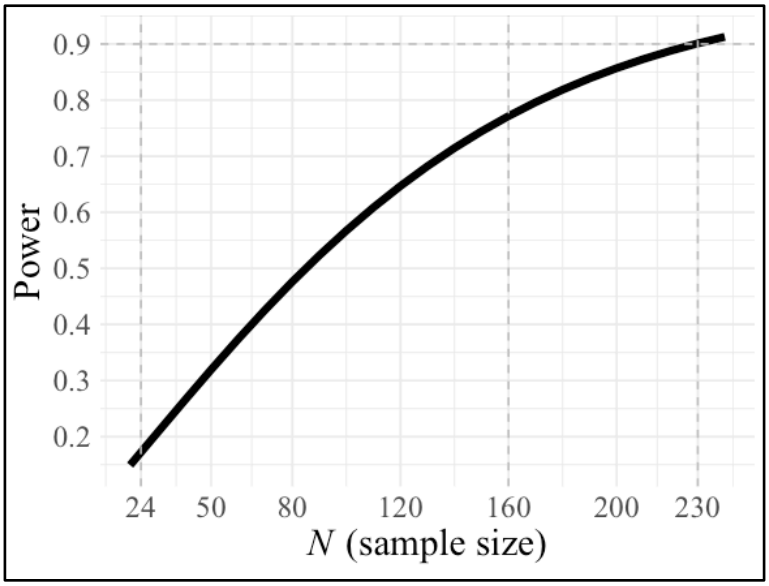
Note. Error bars represent the 95% CI. Panel A includes the complete sample of 159 participants, while Panel B includes a random subsample of 24 participants from the complete sample. In both cases the groups *aware* and *unaware* were almost balanced (81 vs. 78 and 13 vs. 11, respectively).

Further analysis reveals that, for the complete sample, the interaction effect was significant in 78.5% of the simulated samples. While this level of power seems reassuring at first glance, it is based on a very large sample ($N = 159$). More importantly, the power to detect an effect of this size ($\eta_p^2 = .0005$ or $d_s = 0.43$) with a sample equal to the median sample size in this literature (24 participants among all studies reviewed in the Supplemental Material of Vicente-Conesa et al., 2022) is merely 17.21%. This implies that obtaining a non-significant interaction

(as has been consistently observed in previous studies; e.g., Failing et al., 2019; Gao & Theeuwes, 2022; Lin et al., 2021) is actually almost inevitable. If a researcher wants to achieve acceptable statistical power to detect this (plausible) interaction[6], a sample of 229 participants is required to guarantee 90% power. Thus even with 159 participants, Vicente-Conesa et al.'s sample lacks sufficient power to detect a true interaction of the magnitude predicted by the single-process model. Figure 8 illustrates the power curve for sample sizes from 24 to 230.

Figure 8

Statistical power to detect the interaction predicted by the model across a range of sample sizes



Note that the predicted effect size of the interaction is medium-sized across simulations. This is the main reason why larger sample sizes are needed to guarantee sufficient power. However, increasing sample size is not the only means to avoid false-negative conclusions. A small-to-medium-sized observed interaction can also be a symptom that our measures are contaminated by excessive measurement noise, which attenuates the value of the true interaction. To explore to what extent this statistical artifact might be affecting the simulated variables, we estimated their reliability by correlating each variable (suppression score, AW accuracy, and distance from the AW to the HP location) in two different replications. Each violin plot in Figure 9 represents the correlations between 1,000 random pairs of replications. The bottom plots in

[6] These power calculations were obtained with the function `pwr.t.test()` from the R package `{pwr}` (Champely, 2020). Note that, to obtain these numbers, an interaction of 0.43 in d_s units (mean value obtained across simulated samples) was taken as the population effect size and that balanced groups (*aware* and *unaware*) were assumed.

Figure 9 illustrate what we suspected: there is a reliability problem in both suppression scores (Pearson's correlation, M_r = 0.322, sd_r = 0.088) and awareness accuracy (phi coefficient, M_r = 0.145, sd_r = 0.077). For suppression scores, the mean reliability predicted by the model is included within the 95% CI of the reliability estimated in the data, obtained as the permuted split-half reliability across 1,000 replications (M_r = .415 with 95% CI [.200, .597])[7]. For awareness accuracy, a more robust measure of awareness could be derived by using the distances (0-4) from the AW to the HP location (Spearman's correlation, M_r = 0.205, sd_r = 0.075), instead of the binary AW accuracy variable (third plot of Figure 9). Nonetheless, with such measurement uncertainty, again, a non-significant interaction is almost the only empirical scenario a researcher should expect.

What if a researcher obtained half of the measurement noise that we estimate for Vicente-Conesa et al.'s (2022) data? Or a quarter? Further plots in Figure 9 represent these scenarios. We simulated the same original data, but reducing the impact of measurement noise both in RTs (by multiplying the original values of σ_{RT} and τ_{RT} by the square root of 0.50 or 0.25[8]) and in AW (by multiplying Q by 0.50 or 0.25). The result, as expected, is an increase in the measures' reliability, but also in the effect size of the interaction. With half the measurement noise, the interaction increases from d_s = 0.43 to d_s = 0.79, for which 70 participants would be sufficient to achieve 90% power. In turn, with a quarter of the measurement noise, the interaction reaches d_s = 1.38 and only 24 participants (the median sample size in the literature employing the additional singleton task) would finally be sufficient to detect it with 90% power.

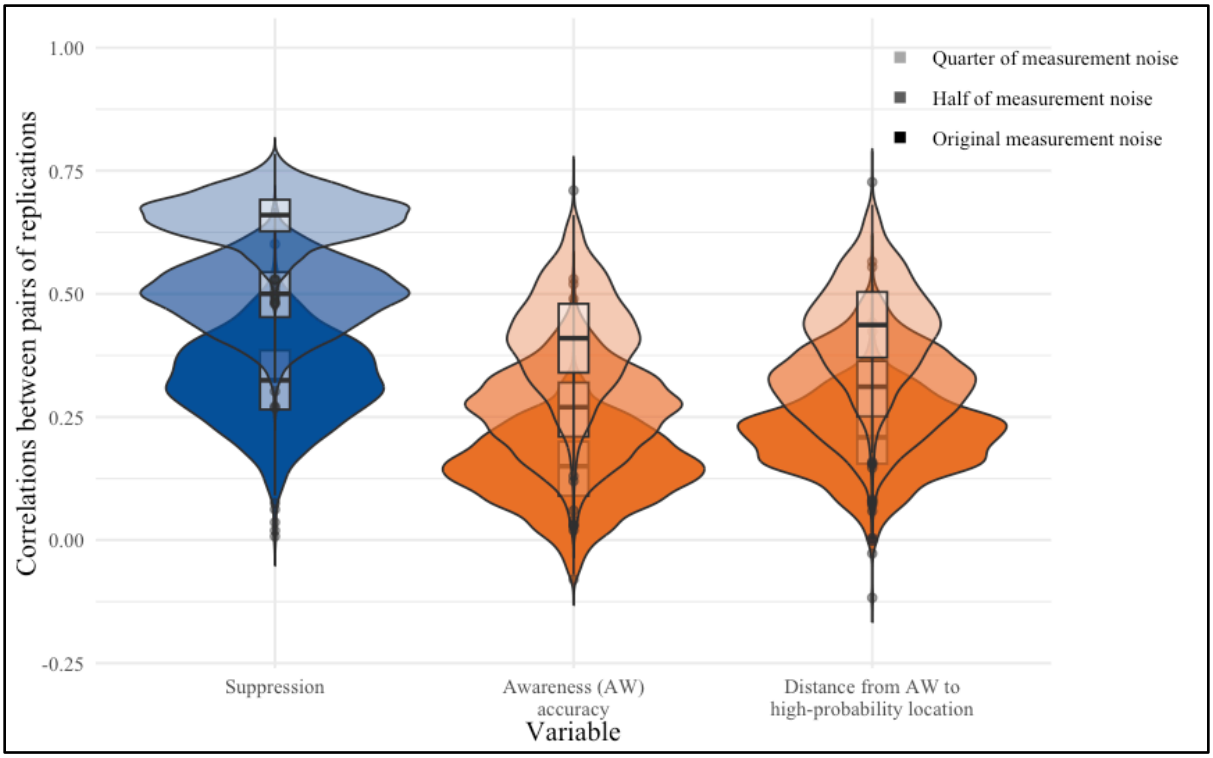
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[7] After applying Spearman-Brown's correction. The R script to compute this coefficient is available at https://osf.io/4qte8/.

[8] The justification for using the square root is to transform the scaling factor (0.50 or 0.25) to the original metric of parameters σ_{RT} and τ_{RT}.

Figure 9

Reliability of simulated variables (suppression, AW accuracy, and distance from the AW response location to the high-probability location, dL_{AW}), computed as the correlation between 1,000 random pairs of replications. The more transparent the violin plots, the lower the measurement noise of the simulated variables



Note. We used Pearson's correlation for suppression scores (continuous variable), the phi coefficient for AW accuracy (binary variable), and Spearman's correlation for dL_{AW} (categorical variable).

Discussion

The modeling approach presented in this work allows researchers to assess the power of inferences in the experimental context in a twofold manner. First, our model allows us to estimate the amount of measurement noise in data collected in the additional singleton task, given a specified level of sampling noise. When these data lead to a non-significant interaction, as in Vicente-Conesa et al.'s (2022) dataset, the model permits us to know if the data are compatible with the theoretical claim that location probability learning and awareness truly interact, but the observed interaction is attenuated by measurement and sampling noise. The model fit of Vicente-Conesa et al.'s dataset is adequate in most of its predictions. However, it suggests that having a large sample size (159 participants is far more than what other studies with the same paradigm have reached) is still insufficient to achieve high statistical power to detect a small-to-moderate

interaction. This is likely due to a serious problem of measurement noise in the suppression and awareness variables, as they have been usually measured.

Second, our model allows us to simulate data for which the correct theoretical model is known, so we can evaluate how sample sizes and reliabilities impact our capacity to infer this model. For the additional singleton task, previous authors have systematically interpreted a null interaction between suppression and awareness as evidence of location probability learning being unconscious. Consistent with Hedge et al. (2018) and LeBel and Paunonen (2011), our results suggest that, even under a model where that interaction is true, the power to detect it rapidly decays as a function of sampling and measurement noise, so obtaining a null interaction becomes a very likely outcome in empirical contexts. This is particularly discouraging for research teams constrained in their ability to access large samples. Instead, they need to obtain measures with little noise to be relatively sure they are able to detect their effects of interest. For instance, one relatively simple way of achieving this is to include more trials per participant. This work also illustrates how easily an effect size can be vulnerable to measurement noise. Hedge et al. (2018) found that even assuming moderate effect sizes, the required sample size for 80% power still exceeds the typically used sample sizes. Consequently, future research will need to exhaustively study how the true size of the intended effect moderates the relationship between measurement/ sampling noise and power.

The modeling approach we adopted opens up alternative methods to estimate reliability in cases where standard methods (e.g., split-half reliability) are inapplicable, as when collecting a single-item measure. The model-based reliability calculated here, computed as the average correlation across pairs of simulated replications, provides a rough estimation in the absence of better evidence. However, we strongly recommend that future designs for the additional singleton task include more informative measures of awareness, if such a construct – and its reliability – is to be properly studied (Adams & Gaspelin, 2020; Cronbach, 1957).

Limitations and future enhancements

Finally, both our model and simulations offer numerous opportunities for enhancement. First, the visual search and awareness model presented here represents just one of several options we considered (see Supplementary Material) and other empirical datasets should be used to gain more evidence on which model fits the data better. Also, it would be profitable to check if other non-linear functions better represent the distances from target and distractor to HP location (e.g.,

an exponential function). Note that here we limited our estimations to the participant level, but other approaches such as multilevel models would offer some interesting additional information (e.g., Rouder & Haaf, 2019, found that their effect estimated from a hierarchical model was more accurate than conventional aggregated estimates). We suspect that, to fulfill the multilevel models' requirements, a Bayesian approach will be more suitable and could overcome potential convergence difficulties (Levy & McNeish, 2023).

We conclude that the heart of the problem in failing to detect an effect does not lie in the reliability of experimental measures themselves. Within the very nature of studying individual differences there is a core concept: between-participants variability. Probably the main scientific task in this field is to evaluate potential relationships between psychological variables. For instance, when studying intelligence or personality traits, no one will doubt that a homogeneous sample would pose a challenge in finding correlates with other variables (i.e., a variable with small variance is unlikely to correlate with others). In contrast, experimental psychology seeks other goals such as finding robust basic effects, prevalent across most individuals within a population. This goal inherently conflicts with variability (Borsboom et al., 2009; Cronbach, 1957) and, when it is low or absent, our measures can neither correlate with others, nor correlate with themselves.

In this latter scenario, a measure that does not correlate with itself entails a measure that cannot be, by definition, reliable. This is a key message: the measure is unreliable, not because it is affected by measurement noise in the population, but because the sample in which it was measured had little variability. The increasingly widespread practice of reporting estimations of reliability is revealing that many common experimental tasks have substantial amounts of measurement noise (Hedge et al., 2018; Paap & Sawi, 2016). However, as Hedge et al. and Rouder et al. (2023a; 2023b) suggest, we are not yet sure whether this implies that our measures need to be enhanced or that their nature in experimental settings makes the classic study of reliability unfeasible.

Some design and statistical artifacts can produce this variability reduction: range restriction, dichotomization, and unbalanced designs. First, if a sample does not represent the range of effects in the whole population, reliability will be underestimated (Fife et al., 2012; Franco-Martínez et al., 2022; Hunter et al., 2006). Since distractor suppression is such a robust effect, it would not be surprising if a lack of participants representing ranges of lower (or even null) suppression is making this variable more unreliable. Second, in the additional singleton task, previous authors

have commonly dichotomized both variables involved in their analysis, losing all the ordinal information the task inherently provides (Hunter & Schmidt, 1990). Authors can gain some variability if they systematically manipulate all levels of distance from the distractor to the HP location, instead of only manipulating the binary variable of whether the distractor appears at the exact HP location or not. The same applies with the awareness measure: the distance from participants' response in the awareness test to the actual HP location is richer in variability than the binary variable of whether they correctly select the HP location or not. Third, another possible source for low empirical reliability would be, as Rouder et al. (2023a, 2023b) called it, excessive trial noise. In Vicente-Conesa et al.'s (2022) design, there are approximately 312 trials for the HP condition but only 168 for the LP condition, per participant. Consequently, this unbalanced design makes trial noise in the latter condition higher than in the former.

Conclusion

In sum, this work provides an illustration that data appearing to offer evidence in favor of a certain theory (i.e., that location probability learning is not related to awareness) may be more consistent with the opposite theory (i.e., that location probability learning and awareness share a common underlying process) if such data are contaminated with measurement and sampling noise. Future simulation studies should explore the influence of alternative design and statistical artifacts (i.e., range restriction, dichotomization, and imbalance) on statistical power. All these, together with the already explored sampling and measurement noise, could provide a wider picture on what obstacles we should avoid when studying individual differences in experimental psychology.

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